Kinematics 1D and 2D Revision Sheet

Level 2 Sheet **Maximum Time Taken to Solve: 50 Min**

Note: Revise Class Notes + Do Last 5 Years of PYQ + These Sheets of Level 1 to 4.

1. A body travels 102.5 m in n^{th} second and 115.0 m in $(n+2)^{\text{th}}$ second. The acceleration is :

(A) 6.25 m/s^2 (B) 12.5 m/s^2

(C) 9 m/s^2 (D) 5 m/s^2

2. A paratrooper jumps from an aeroplane and opens a parachute after 2 s of free fall and starts deaccelerating with 3 m/s^2. At 10 m height from ground, while descending with the help of parachute, the speed of paratrooper is 5 m/s. The initial height of the airplane is _____ m. ($g = 10$ m/s^2)

(A) 82.5 (B) 20

(C) 62.5 (D) 92.5

3. If the velocity of a body related to displacement x is given by $v = \sqrt{5000 + 24x}$ m s^{-1}, then acceleration of the body is _____ m s^{-2}.

4. A particle starts moving from time $t = 0$ and its coordinate is given as $x(t) = 4t^3 - 3t$

A. The particle returns to its original position (origin) 0.866 units later

B. The particle is 1 unit away from origin at its turning point

C. Acceleration of the particle is non-negative

D. The particle is 0.5 units away from origin at its turning point

E. Particle never turns back as acceleration is non-negative

Choose the correct answer from the options given below :

(A) A, B, C Only (B) C, E Only

(C) A, C Only (D) A, C, D Only

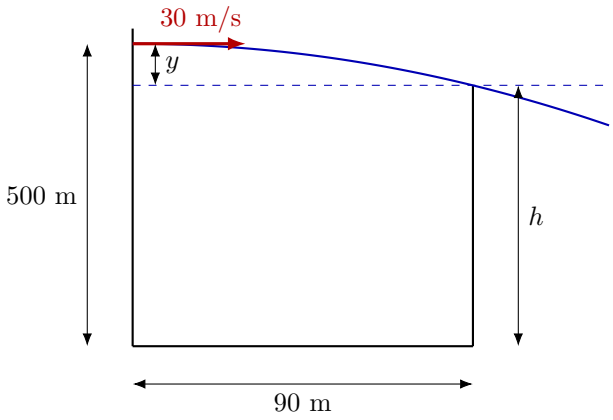
5. Two cars are travelling towards each other at speed of 20 m s^{-1} each. When the cars are 300 m apart, both the drivers apply brakes and the cars retard at the rate of 2 m s^{-2}. The distance between them when they come to rest is :

(A) 200 m (B) 100 m

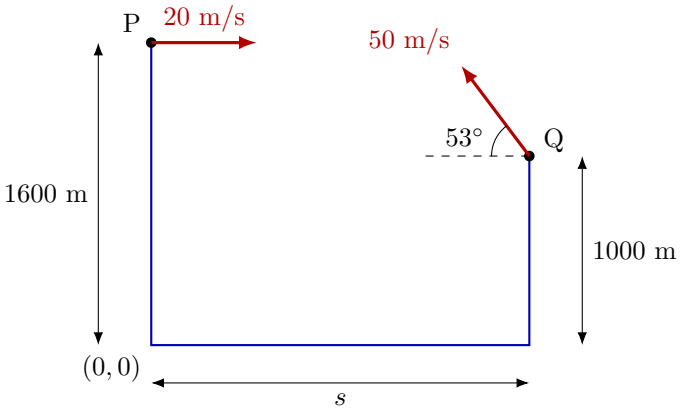
(C) 50 m (D) 25 m

6. The maximum speed of a boat in still water is 27 km/h. Now this boat is moving downstream in a river flowing at 9 km/h. A man in the boat throws a ball vertically upwards with speed of 10 m/s. Range of the ball as observed by an observer at rest on the river bank, is _____ cm. (Take $g = 10$ m/s^2)

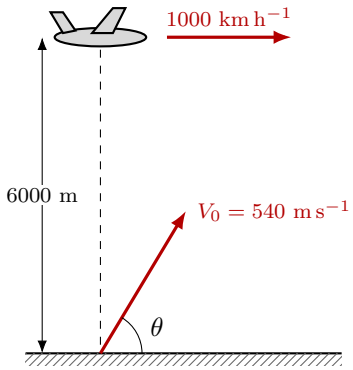
7. Find 'h' if projectile is just grazing the wall ?



8. Both are thrown simultaneously. If they collide in mid air, find s ?



9. An aircraft moving with a speed of 1000 $\mathrm{km\,h^{-1}}$ is at a height of 6000 m, just overhead of an anti-aircraft gun. If the muzzle velocity is 540 $\mathrm{m\,s^{-1}}$, the firing angle θ should be



(a) 73° (b) 30° (c) 59° (d) 45°

10. A particle is projected at 60° to the horizontal with a kinetic energy K. The kinetic energy at the highest point is

(a) K (b) Zero (c) $K/2$ (d) $K/4$

11. A particle is projected from a point O with velocity u at an angle of 60° with the horizontal. When it is moving in a direction at right angles to its direction at O, its velocity then is given by

(a) $\frac{u}{3}$ (b) $\frac{u}{2}$ (c) $\frac{2u}{3}$ (d) $\frac{u}{\sqrt{3}}$

12. Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R. If the time of flight of these bodies are 5 s and 10 s, respectively, then the value of R is _____ m. (Take $g = 10$ m/s^2)

(A) 250 (B) 25

(C) 500 (D) 125

13. A projectile is thrown upward at an angle 60° with the horizontal. The speed of the projectile is 20 m/s when its direction of motion is 45° with the horizontal. The initial speed of the projectile is _____ m/s.

(A) $20\sqrt{2}$ (B) 40

(C) $40\sqrt{2}$ (D) $20\sqrt{3}$

Keep Going!

"It does not matter how slowly you go as long as you do not stop."

— Confucius